Course PH410 Course coordinator: Alika Khare

Problem Set _05 Due Date: Feb 17, 2025

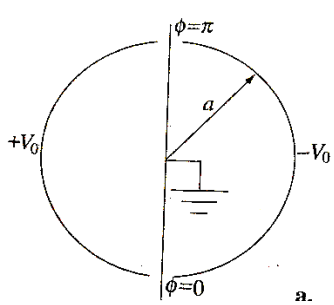
Note: Please submit the solutions of at least one of the problems, on Monday, Feb 17, 2025, during the class. The solutions can be submitted in a group of maximum of four students. Name and Roll no. of all the group members should be written clearly on the top. If solutions are found to be reasonably correct then a total of 05 marks will be awarded and in case of non-submission/wrong solution zero marks. List all the boundary conditions as applicable. Where ever applicable draw the well labeled diagram along with axis properly marked.

1. The curved surface of an infinitely long cylindrical shell, of radius R, carries a surface charge density $\sigma(\varphi) = k\, sin5\varphi$ (where k is a constant).
   a. Find the potential inside as well outside the cylinder.
   b. From above work out the electric field for $\rho>R$ as well as $\rho<R$.

2. A cylindrical shell of infinite length contains a uniform surface charge density σ_o on its one half of surface and that of second half surface $-\sigma_o$. Obtain the potential inside and outside the surface.

3. The curved surface of a cylindrical shaped conductor of infinite length has been divided into two equal haves. The first part having, polar angle φ in the range of $0 \leq \varphi < \pi$, is maintained at a potential of V_1 and that of the remaining part at V_2. Obtain the expression of the potential inside as well outside. Cross check your result back at the surface.

4. The principal of electrostatic precipitator is based on the above problem. It is used to collect all the charged particles from the ashes. The schematic of the top view of one such electrostatic precipitator is shown the diagram below. The central conductor is in the form of a mesh to allow the passage of the particles (it doesn't affect the solution in the present case). Calculate the potential inside and outside using above results.

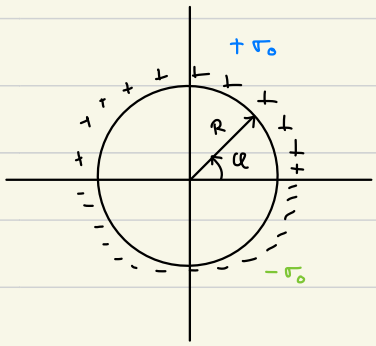


a.

5. A long metallic cylinder of radius R is divided into four equal sections section (length wise). Three of these are grounded and fourth one (e.g. $-\pi/4<\varphi<\pi/4$) insulated from rest is maintained at a constant potential V_o. Obtained the expression for induced surface charge density and hence the line charge density on the section opposite to the V_o..

2. A cylindrical shell of infinite length contains a uniform surface charge density σ_o on its one half of surface and that of second half surface $-\sigma_o$. Obtain the potential inside and outside the surface.

$z \rightarrow$ symmetry



$$V(\rho,\varphi) = A_0 + B_0 \ln \rho + \sum_{n=1}^{\infty} \left[A_n \rho^n + \frac{B_n}{\rho^n}\right]\left[C_n \cos(n\varphi) + D_n \sin(n\varphi)\right]$$

$\rho \rightarrow \infty$ then $V(\rho,\varphi) = 0 \Rightarrow A_0 = 0$

$\rho < R \Rightarrow B_0 = 0$ and $B_n = 0$

$$V_{inside} = \sum_{n=1}^{\infty} \rho^n \left(E_n \cos(n\varphi) + F_n \sin(n\varphi)\right)$$

$\rho \geq R \Rightarrow B_0 = 0$ and $A_n = 0$

$$V_{outside} = \sum_{n=1}^{\infty} \rho^{-n} \left(G_n \cos(n\varphi) + H_n \sin(n\varphi)\right)$$

Continuity of $V(\rho = R)$

$$\sum_{n=1}^{\infty} R^n \left(E_n \cos(n\varphi) + F_n \sin(n\varphi)\right) = \sum_{n=1}^{\infty} R^{-n} \left(G_n \cos(n\varphi) + H_n \sin(n\varphi)\right)$$

$$G_n = R^{2n} E_n$$

$$H_n = R^{2n} F_n$$

$$\left.\frac{\partial V_{out}}{\partial \rho}\right|_{\rho=R} - \left.\frac{\partial V_{ins}}{\partial \rho}\right|_{\rho=R} = -\frac{\sigma}{\varepsilon_0}$$

$$\sum_{n=1}^{\infty} \frac{-n}{R^{n+1}} R^{2n} \left(E_n \cos n\varphi + F_n \sin n\varphi\right)$$

$$- \sum_{n=1}^{\infty} n R^{n-1} \left(E_n \cos n\varphi + F_n \sin n\varphi\right) = -\frac{\sigma}{\varepsilon_0}$$

$$\sum_{n=1}^{\infty} 2n R^{n-1} \left(E_n \cos n\varphi + F_n \sin n\varphi\right) = \frac{\sigma}{\varepsilon_0}$$

$$\sum_{n=1}^{\infty} 2n R^{n-1} \left[E_n \int_0^{2\pi} \cos n\varphi \cos l\varphi \, d\varphi + F_n \int_0^{2\pi} \sin n\varphi \cos l\varphi \, d\varphi \right] = \int_0^{2\pi} \frac{\sigma}{\varepsilon_0} \cos l\varphi \, d\varphi$$

$$2l R^{l-1} E_l \pi = \frac{\sigma_0}{\varepsilon_0} \left[\int_0^{\pi} \cos l\varphi \, d\varphi - \int_{\pi}^{2\pi} \cos l\varphi \, d\varphi \right]$$

$$= 0$$

$$E_l = 0$$

$$\sum_{n=1}^{\infty} 2l\, R^{n-1}\left[E_n \int_0^{2\pi} \cos n\varphi \sin l\varphi \, d\varphi + F_n \int_0^{2\pi} \sin n\varphi \sin l\varphi \, d\varphi\right] = \int_0^{2\pi} \frac{\sigma}{\varepsilon_0} \sin l\varphi \, d\varphi$$

$$2l\, R^{l-1} F_l \pi = \frac{\sigma_0}{\varepsilon_0}\left[\int_0^{\pi} \sin l\varphi \, d\varphi - \int_{\pi}^{2\pi} \sin l\varphi \, d\varphi\right]$$

$$2l\, R^{l-1} F_l \pi = \frac{\sigma_0}{l\varepsilon_0}\left[\cos l\varphi \Big|_{\pi}^{0} - \cos l\varphi \Big|_{2\pi}^{\pi}\right.$$

$$2l\, R^{l-1} F_l \pi = \frac{\sigma_0}{l\varepsilon_0}\left[1 - \cos \pi l - \cos \pi l + 1\right]$$

$$2l\, R^{l-1} F_l \pi = \frac{\sigma_0}{l\varepsilon_0}\left[2 - 2\cos(\pi l)\right]$$

$$F_l = \frac{\sigma_0}{l^2 \varepsilon_0 \pi R^{l-1}}\left[1 - \cos(\pi l)\right]$$

$$F_l = \begin{cases} 0 & , \quad l = 0, 2, 4, \ldots \\ \dfrac{2\sigma_0}{l^2 \varepsilon_0 \pi R^{l-1}} & , \quad l = 1, 3, 5, \ldots \end{cases}$$

$$V_{inside} = \sum_{l=1,3,5,\ldots}^{\infty} \rho^{l} \left[\frac{2\sigma_0}{l^2 \varepsilon_0 \pi R^{l-1}} \sin(l\varphi) \right]$$

$$= \frac{2\sigma_0 R}{\varepsilon_0 \pi} \sum_{l=1,3,5,\ldots}^{\infty} \frac{\rho^{l}}{l^2 R^{l}} \sin(l\varphi)$$

$$V_{outside} = \sum_{l=1,3,5,\ldots}^{\infty} \rho^{-l} R^{2l} \left[\frac{2\sigma_0}{l^2 \varepsilon_0 \pi R^{l-1}} \sin(l\varphi) \right]$$

$$= \frac{2\sigma_0 R}{\varepsilon_0 \pi} \sum_{l=1,3,5,\ldots}^{\infty} \frac{\rho^{-l} R^{l}}{l^2} \sin(l\varphi)$$

$$V(\rho,\varphi) = \frac{2\sigma_0 R}{\varepsilon_0 \pi} \sum_{l=1,3,5,\ldots}^{\infty} \frac{\sin(l\varphi)}{l^2} \begin{cases} (\rho/R)^{l} & \rho < R \\ (R/\rho)^{l} & \rho > R \end{cases}$$

3. The curved surface of a cylindrical shaped conductor of infinite length has been divided into two equal haves. The first part having, polar angle φ in the range of $0 \leq \varphi < \pi$, is maintained at a potential of V_1 and that of the remaining part at V_2. Obtain the expression of the potential inside as well outside. Cross check your result back at the surface.

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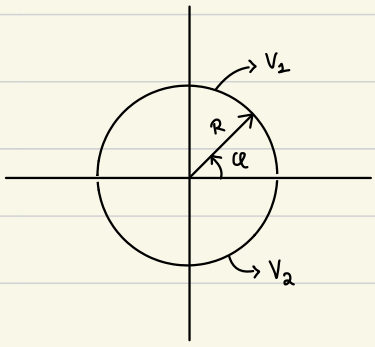

$$V(\rho,\varphi) = A_0 + B_0 \ln \rho + \sum_{n=1}^{\infty} \left[A_n \rho^n + \frac{B_n}{\rho^n}\right]\left[C_n \cos(n\varphi) + D_n \sin(n\varphi)\right]$$

$\rho \leq R$

$$V_{ins}(\rho,\varphi) = A_0 + \sum_{n=1}^{\infty} \rho^n \left[C_n \cos(n\varphi) + D_n \sin(n\varphi)\right]$$

$\rho \geq R$

$$V_{out}(\rho,\varphi) = B_0 + \sum_{n=1}^{\infty} \rho^{-n} \left[E_n \cos(n\varphi) + F_n \sin(n\varphi)\right]$$

$$A_0 + \sum_{n=1}^{\infty} R^n \left[C_n \cos(n\varphi) + D_n \sin(n\varphi)\right] = B_0 + \sum_{n=1}^{\infty} R^{-n} \left[E_n \cos(n\varphi) + F_n \sin(n\varphi)\right]$$

$$E_n = R^{2n} C_n$$
$$F_n = R^{2n} D_n$$
$$A_0 = B_0$$

$$A_0 + \sum_{n=1}^{\infty} \rho^n \left[C_n \cos(n\varphi) + D_n \sin(n\varphi) \right] = V_{ins}$$

$$\rightarrow \int_0^{2\pi} A_0 \, d\varphi + \sum_{n=1}^{\infty} R^n \left[C_n \int_0^{2\pi} \cos(n\varphi) d\varphi + D_n \int_0^{2\pi} \sin(n\varphi) d\varphi \right] = \int_0^{\pi} V_1 \, d\varphi + \int_0^{\pi} V_2 \, d\varphi$$

$$A_0 \, 2\pi = (V_1 + V_2)\pi$$

$$A_0 = \frac{V_1 + V_2}{2}$$

$$\rightarrow \int_0^{2\pi} A_0 \sin(l\varphi) d\varphi + \sum_{n=1}^{\infty} R^n \left[C_n \int_0^{2\pi} \cos(n\varphi) \sin(l\varphi) \, d\varphi \right.$$

$$\left. + D_n \int_0^{2\pi} \sin(n\varphi) \sin(l\varphi) \, d\varphi \right] = \int_0^{\pi} V_1 \sin(l\varphi) \, d\varphi + \int_{\pi}^{2\pi} V_2 \sin(l\varphi) \, d\varphi$$

$$R^n D_n \pi = \frac{[1 - (-1)^n]}{n} (V_1 - V_2)$$

$$D_n = \frac{2(V_1 - V_2)}{R^n n \pi} \quad [\text{odd } n]$$

$$\rightarrow \int_0^{2\pi} A_0 \cos(l\varphi)\, d\varphi + \sum_{n=1}^{\infty} R^n \left[C_n \int_0^{2\pi} \cos(n\varphi)\cos(l\varphi)\, d\varphi \right.$$

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$$R^n C_n \pi = 0$$

$$C_n = 0$$

$$V_{ins}(\rho,\varphi) = \left(\frac{V_1+V_2}{2}\right) + \sum_{n=1,3,5,\dots}^{\infty} \frac{2(V_1-V_2)}{n\pi} \left(\frac{\rho}{R}\right)^n \sin(n\varphi)$$

$\rightarrow$ We know,

$$E_n = R^{2n} C_n$$
$$F_n = R^{2n} D_n$$
$$A_0 = B_0$$

$$E_n = 0$$

$$F_n = \frac{2(V_1-V_2)}{R^n\, n\pi} R^{2n} = \frac{2(V_1-V_2)}{n\pi} R^n$$

$$B_0 = \frac{V_1+V_2}{2}$$

$$V_{out}(\rho,\varphi) = \left(\frac{V_1+V_2}{2}\right) + \sum_{n=1,3,5,\dots}^{\infty} \frac{2(V_1-V_2)}{n\pi} \left(\frac{R}{\rho}\right)^n \sin(n\varphi)$$

4. The principal of electrostatic precipitator is based on the above problem. It is used to collect all the charged particles from the ashes. The schematic of the top view of one such electrostatic precipitator is shown the diagram below. The central conductor is in the form of a mesh to allow the passage of the particles (it doesn't affect the solution in the present case). Calculate the potential inside and outside using above results.

$$V_{ins}(\rho,\varphi) = \left(\frac{V_1+V_2}{2}\right) + \sum_{n=1,3,5,\ldots}^{\infty} \frac{2(V_1-V_2)}{n\pi}\left(\frac{\rho}{R}\right)^n \sin(n\varphi)$$

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$$= \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_0}{n\pi}\left(\frac{\rho}{R}\right)^n \sin(n\varphi)$$

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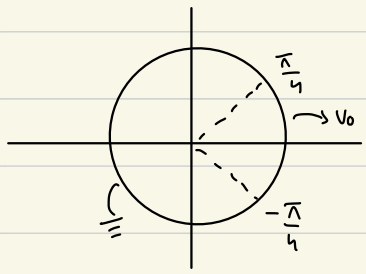
5. A long metallic cylinder of radius R is divided into four equal sections section (length wise). Three of these are grounded and fourth one (e.g. $-\pi/4<\varphi<\pi/4$) insulated from rest is maintained at a constant potential V_o. Obtained the expression for induced surface charge density and hence the line charge density on the section opposite to the V_o..



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$$\rightarrow \int_0^{2\pi} A_0 \, d\varphi + \sum_{n=1}^{\infty} R^n \left[C_n \int_0^{2\pi} \cos(n\varphi) \, d\varphi + D_n \int_0^{2\pi} \sin(n\varphi) \, d\varphi \right] = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} V_0 \, d\varphi$$

$$A_0 \, 2\pi = V_0 \frac{\pi}{2}$$

$$A_0 = \frac{V_0}{4}$$

$$\rightarrow \int_0^{2\pi} A_0 \sin(l\varphi) \, d\varphi + \sum_{n=1}^{\infty} R^n \left[C_n \int_0^{2\pi} \cos(n\varphi) \sin(l\varphi) \, d\varphi \right.$$

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$$R^n D_n \pi = V_0 \left. \frac{\cos n\varphi}{n} \right|_{\frac{\pi}{4}}^{-\frac{\pi}{4}}$$

$$D_n = 0$$

$$\rightarrow \int_0^{2\pi} A_0 \cos(l\varphi) \, d\varphi + \sum_{n=1}^{\infty} R^n \left[C_n \int_0^{2\pi} \cos(n\varphi) \cos(l\varphi) \, d\varphi \right.$$

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$$R^n C_n \pi = V_0 \left. \frac{\sin(n\varphi)}{n} \right|_{-\frac{\pi}{4}}^{\pi/4}$$

$$R^n C_n \pi = \frac{V_0}{n} 2 \sin\left(n \frac{\pi}{4}\right)$$

$$C_n = \frac{2V_0}{n\pi} \frac{1}{R^n} \sin\left(n\frac{\pi}{4}\right)$$

$$V_{ins}(\rho, \varphi) = \frac{V_0}{4} + \sum_{n=1}^{\infty} \frac{2V_0}{n\pi} \sin\left(n\frac{\pi}{4}\right) \left(\frac{\rho}{R}\right)^n \cos(n\varphi)$$

$$V_{out}(\rho, \varphi) = \frac{V_0}{4} + \sum_{n=1}^{\infty} \frac{2V_0}{n\pi} \sin\left(n\frac{\pi}{4}\right) \left(\frac{R}{\rho}\right)^n \cos(n\varphi)$$

1. The curved surface of an infinitely long cylindrical shell, of radius R, carries a surface charge density $\sigma(\varphi) = k \sin 5\varphi$ (where k is a constant).
   a. Find the potential inside as well outside the cylinder.
   b. From above work out the electric field for $\rho > R$ as well as $\rho < R$.

$$V(\rho, \varphi) = A_0 + B_0 \ln \rho + \sum_{n=1}^{\infty} \left[A_n \rho^n + \frac{B_n}{\rho^n} \right] \left[C_n \cos(n\varphi) + D_n \sin(n\varphi) \right]$$

$\rho \leq R$

$$V_1(\rho, \varphi) = \sum_{n=1}^{\infty} \rho^n \left[C_n \cos(n\varphi) + D_n \sin(n\varphi) \right]$$

$\rho \geq R$

$$V_2(\rho, \varphi) = \sum_{n=1}^{\infty} \rho^{-n} \left[E_n \cos(n\varphi) + F_n \sin(n\varphi) \right]$$

$$\left. \frac{\partial V_2}{\partial \rho} \right|_{r=R} - \left. \frac{\partial V_1}{\partial \rho} \right|_{r=R} = -\frac{\sigma}{\varepsilon_0}$$

$$\sum_{n=1}^{\infty} R^n \left[C_n \cos(n\varphi) + D_n \sin(n\varphi) \right] = \sum_{n=1}^{\infty} R^{-n} \left[E_n \cos(n\varphi) + F_n \sin(n\varphi) \right]$$

$$E_n = R^{2n} C_n$$
$$F_n = R^{2n} D_n$$

$$\sum_{n=1}^{\infty} (-n) R^{-n-1} \left[E_n \cos(n\varphi) + F_n \sin(n\varphi) \right]$$

$$- \sum_{n=1}^{\infty} n R^{n-1} \left[C_n \cos(n\varphi) + D_n \sin(n\varphi) \right] = -\frac{\sigma}{\varepsilon_0}$$

$$\sum_{n=1}^{\infty} (-n)\, R^{-n-1} \left[R^{2n} C_n \cos(n\varphi) + R^{2n} D_n \sin(n\varphi) \right]$$

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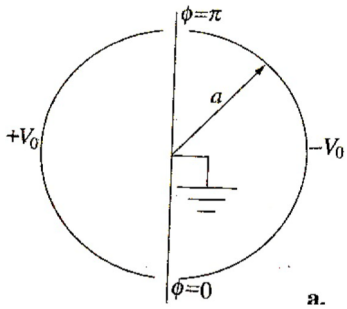
$$- \sum_{n=1}^{\infty} 2 n R^{n-1} \left[C_n \cos(n\varphi) + D_n \sin(n\varphi) \right] = - \frac{\sigma}{\varepsilon_0}$$

4. The principal of electrostatic precipitator is based on the above problem. It is used to collect all the charged particles from the ashes. The schematic of the top view of one such electrostatic precipitator is shown the diagram below. The central conductor is in the form of a mesh to allow the passage of the particles (it doesn't affect the solution in the present case). Calculate the potential inside and outside using above results.



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